

Exercise 7: Packages (MySQL Equivalent)

Aim

To organize related procedures and functions into logical modules (similar to Oracle Packages) for customer, employee, and account management.

Scenario 1: Customer Management Package

--**Question:** Create a package **CustomerManagement** with procedures for adding a new customer, updating customer details, and a function to get customer balance.

-- **Procedure 1: Add Customer**

DELIMITER \$\$

CREATE PROCEDURE AddCustomer(

IN CID INT,

IN CName VARCHAR(50),

IN CAge INT,

IN CBalance DECIMAL(10,2),

IN Interest DECIMAL(5,2)

)

BEGIN

INSERT INTO Customer

(CustomerID, CustomerName, Age, Balance, LoanInterestRate, IsVIP)

VALUES

(CID,CName,CAge,CBalance,Interest,FALSE);

END \$\$

DELIMITER ;

```
SELECT * FROM Customer;
```

[illegible]

```
CALL AddCustomer(107,' Anshita ',20,18000,9.5);
```

```
SELECT * FROM Customer;
```

[illegible]

--Procedure 2: Update Customer

DELIMITER \$\$

CREATE PROCEDURE UpdateCustomerDetails(

IN CID INT,

IN NewBalance DECIMAL(10,2)

)

BEGIN

UPDATE Customer

SET Balance=NewBalance

WHERE CustomerID=CID;

END \$\$

DELIMITER ;

SELECT * FROM Customer;

	CustomerID	CustomerName	Age	Balance	LoanInterestRate	IsVIP	DOB	LastModified
▶	101	Krishna	65	25000.00	2.00	1	1961-05-15	2026-06-30
	102	Rahul	45	8000.00	9.50	1	1981-09-20	2026-06-30
	103	Priya	70	20000.00	1.00	1	1956-12-10	2026-06-30
	104	Arun	35	12000.00	10.50	1	1991-07-08	2026-06-30
	105	Sneha	62	5000.00	1.50	0	1964-03-25	2026-06-30
	106	Karan	30	12000.00	9.00	0	NULL	2026-06-30
	107	Anshita	20	18000.00	9.50	0	NULL	NULL
	112	Karan	28	18000.00	9.50	0	NULL	NULL
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

CALL UpdateCustomerDetails(101,30000);

SELECT * FROM Customer;

	CustomerID	CustomerName	Age	Balance	LoanInterestRate	IsVIP	DOB	LastModified
▶	101	Krishna	65	30000.00	2.00	1	1961-05-15	2026-06-30
	102	Rahul	45	8000.00	9.50	1	1981-09-20	2026-06-30
	103	Priya	70	20000.00	1.00	1	1956-12-10	2026-06-30
	104	Arun	35	12000.00	10.50	1	1991-07-08	2026-06-30
	105	Sneha	62	5000.00	1.50	0	1964-03-25	2026-06-30
	106	Karan	30	12000.00	9.00	0	NULL	2026-06-30
	107	Anshita	20	18000.00	9.50	0	NULL	NULL
	112	Karan	28	18000.00	9.50	0	NULL	NULL
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

--Function 3: Get Customer Balance

DELIMITER \$\$

CREATE FUNCTION GetCustomerBalance(

CID INT

)

RETURNS DECIMAL(10,2)

DETERMINISTIC

BEGIN

DECLARE Bal DECIMAL(10,2);

SELECT Balance

INTO Bal

FROM Customer

WHERE CustomerID=CID;

RETURN Bal;

END \$\$

DELIMITER ;

SELECT GetCustomerBalance(101);

	GetCustomerBalance(101)
▶	30000.00

Scenario 2: Employee Management Package

--**Question:** Write a package **EmployeeManagement** with procedures to hire new employees, update employee details, and a function to calculate annual salary.

--**Procedure 1: Hire Employee**

DELIMITER \$\$

CREATE PROCEDURE HireEmployee(

IN EID INT,

IN EName VARCHAR(50),

IN Salary DECIMAL(10,2),

IN Dept VARCHAR(30)

)

BEGIN

INSERT INTO Employee

VALUES

(EID,EName,Salary,Dept);

END \$\$

DELIMITER ;

SELECT * FROM EMPLOYEE;

	EmployeeID	EmployeeName	Salary	Department
▶	201	Ravi	60000.00	HR
	202	Anita	66000.00	Finance
	203	John	49500.00	Finance
✱	NULL	NULL	NULL	NULL

```
CALL HireEmployee(204,'Vishwa',55000,'HR');
```

```
SELECT * FROM EMPLOYEE;
```

	EmployeeID	EmployeeName	Salary	Department
▶	200	Vishwa	55000.00	HR
	201	Ravi	60000.00	HR
	202	Anita	66000.00	Finance
	203	John	49500.00	Finance
✱	NULL	NULL	NULL	NULL

--Procedure 2: Update Employee

```
DELIMITER $$
```

```
CREATE PROCEDURE UpdateEmployeeDetails(
```

```
IN EID INT,
```

```
IN NewSalary DECIMAL(10,2)
```

```
)
```

```
BEGIN
```

```
UPDATE Employee
```

```
SET Salary=NewSalary
```

```
WHERE EmployeeID=EID;
```

```
END $$
```

```
DELIMITER ;
```

```
SELECT * FROM Employee;
```

	EmployeeID	EmployeeName	Salary	Department
▶	200	Vishwa	55000.00	HR
	201	Ravi	60000.00	HR
	202	Anita	66000.00	Finance
	203	John	49500.00	Finance
✱	NULL	NULL	NULL	NULL

```
CALL UpdateEmployeeDetails(201,65000);
```

```
SELECT * FROM Employee;
```

	EmployeeID	EmployeeName	Salary	Department
▶	200	Vishwa	55000.00	HR
	201	Ravi	65000.00	HR
	202	Anita	66000.00	Finance
	203	John	49500.00	Finance
•	NULL	NULL	NULL	NULL

--Function 3: Calculate Annual Salary

```
DELIMITER $$
```

```
CREATE FUNCTION CalculateAnnualSalary(
```

```
MonthlySalary DECIMAL(10,2)
```

```
)
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
RETURN MonthlySalary*12;
```

```
END $$
```

```
DELIMITER ;
```

```
SELECT CalculateAnnualSalary(60000);
```

	CalculateAnnualSalary(60000)
▶	720000.00

Scenario 3: Account Operations Package

--**Question:** Create a package **AccountOperations** with procedures for opening a new account, closing an account, and a function to get the total balance of a customer across all accounts.

--**Procedure 1: Open Account**

DELIMITER \$\$

CREATE PROCEDURE OpenAccount(

IN AccID INT,

IN CID INT,

IN AccType VARCHAR(20),

IN Bal DECIMAL(10,2)

)

BEGIN

INSERT INTO Account

VALUES

(AccID,CID,AccType,Bal);

END \$\$

DELIMITER ;

SELECT * FROM Account;

	AccountID	CustomerID	AccountType	Balance
▶	1	101	Savings	45300.00
	2	102	Savings	35100.00
	3	103	Savings	14950.00
	4	104	Current	40200.00
	5	105	Savings	25050.00
•	NULL	NULL	NULL	NULL

```
CALL OpenAccount(6,101,'Savings',15000);
SELECT * FROM Account;
```

	AccountID	CustomerID	AccountType	Balance
▶	1	101	Savings	45300.00
	2	102	Savings	35100.00
	3	103	Savings	14950.00
	4	104	Current	40200.00
	5	105	Savings	25050.00
	6	101	Savings	15000.00
•				

--Procedure 2: Close Account

```
DELIMITER $$
```

```
CREATE PROCEDURE CloseAccount(
```

```
IN AccID INT
```

```
)
```

```
BEGIN
```

```
DELETE FROM Account
```

```
WHERE AccountID=AccID;
```

```
END $$
```

```
DELIMITER ;
```

```
SELECT * FROM Account;
```

	AccountID	CustomerID	AccountType	Balance
▶	1	101	Savings	45300.00
	2	102	Savings	35100.00
	3	103	Savings	14950.00
	4	104	Current	40200.00
	5	105	Savings	25050.00
	6	101	Savings	15000.00
•	NULL	NULL	NULL	NULL

```
CALL CloseAccount(6);
```

```
CALL CloseAccount(5);
```

```
SELECT * FROM Account;
```

	AccountID	CustomerID	AccountType	Balance
▶	1	101	Savings	45300.00
	2	102	Savings	35100.00
	3	103	Savings	14950.00
	4	104	Current	40200.00
★	NULL	NULL	NULL	NULL

--Function 3: Get Total Balance

```
DELIMITER $$
```

```
CREATE FUNCTION GetTotalBalance(
```

```
CID INT)
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
DECLARE Total DECIMAL(10,2);
```

```
SELECT SUM(Balance)
```

```
INTO Total
```

```
FROM Account
```

```
WHERE CustomerID=CID;
```

```
RETURN IFNULL(Total,0);
```

```
END $$
```

```
DELIMITER ;
```

```
SELECT GetTotalBalance(101);
```

	GetTotalBalance(101)
▶	45300.00